PROJECT PROPOSAL REPORT

SMART SOLAR POWERED PADDY DRYER

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1.PROJECT OVERVIEW

1.1 INTRODUCTION

Drying paddy is an essential post-harvest process required to reduce the moisture content of the grain from about 20–25% down to a safe storage level of 12–14%. In countries like Sri Lanka, farmers mostly rely on traditional open sun drying methods. However, this method has several major drawbacks. It is highly dependent on unpredictable weather, takes a long time, and exposes the harvest to dust, birds, and insects. Furthermore, open sun drying often leads to uneven drying, which causes the rice grains to crack during milling, ultimately resulting in high post-harvest losses and lower market value.

To address these critical issues, this project introduces the design of a Smart Solar Powered Paddy Dryer. This system offers an efficient, off grid solution by utilizing solar energy not only to heat the drying air but also to power the electrical components. The system forces the solar heated air through a drying chamber to remove moisture from the paddy continuously. By combining concepts from mechanical, thermal, and control engineering, this project aims to provide a reliable, low cost, and energy efficient solution that significantly improves product quality and reduces post harvest losses for local farming communities.

1.2 PURPOSE AND NEEDS

Fresh paddy is generally harvested at a high moisture content ranging from 20% to 27% on a wet basis to prevent grain shattering in the field. However, to ensure proper milling and long-term storage, this moisture level must be rapidly reduced to a safe threshold of 14%, ideally within 24 hours of harvest. Failure to achieve this critical moisture reduction leads to severe degradation, including grain discoloration, the promotion of mold development, and an increased susceptibility to pest attacks. Consequently, inefficient post-harvest operations account for massive agricultural deficits; globally, one-third of all food produced is lost annually, with approximately 75% of total post-harvest losses in major food grains like rice occurring directly at the farm level.

To combat these substantial agricultural deficits and ensure food security, there is a critical and immediate need for an innovative, highly controlled drying solution. The proposed off-grid, smart solar-powered paddy dryer fulfills this necessity. By providing a scientifically managed, weather-independent drying environment, this system ensures rapid and precise moisture reduction. It entirely removes the guesswork from post-harvest operations while operating without heavy reliance on unstable grid electricity or expensive fossil fuels. Ultimately, this project aims to empower rural agricultural communities with accessible, smart technology that preserves grain quality, maximizes market value, and significantly mitigates post-harvest crop losses.

2.ANALYSIS OF EXISTING SYSTEMS

2.1 SPECIFICATIONS

To design an optimized paddy dryer, it is essential to first understand the operational mechanics of current drying systems used in the agricultural sector. The most prevalent method among smallholder farmers is traditional open sun drying, which involves spreading harvested wet paddy in thin layers across open surfaces, such as concrete floors or woven mats. This natural process relies entirely on ambient solar radiation to heat the grain and natural wind currents to carry away evaporated moisture, requiring continuous manual turning to expose all sides of the grain to the sun. Alternatively, conventional mechanical and industrial dryers operate by utilizing heavy-duty electric heating coils or fossil-fuel (diesel/gas) burners to generate high temperatures. These systems use high-capacity industrial blower fans to force the heated air through large drying chambers, offering a highly controlled environment designed to process massive grain capacities rapidly. Bridging the gap between these two are passive solar dryers, which enclose the grain within a transparent structure (like a greenhouse). These rely on the greenhouse effect to trap solar radiation and passively raise the internal temperature, utilizing natural thermal drafts rather than mechanical fans to move the air.

2.2 LIMITATION

Despite their widespread application, each of these existing systems possesses critical limitations that significantly hinder efficient post harvest operations. Traditional open sun drying is severely constrained by its absolute dependency on clear weather conditions. Sudden rainfall or prolonged cloud cover immediately halts the drying process, leading to rapid grain degradation, severe contamination from pests and particulate matter, and uneven drying that causes grains to crack during the milling process. Conversely, while mechanical dryers resolve weather dependency, they are highly energy intensive due to the significant latent heat required for water evaporation. The exorbitant initial capital required, coupled with high continuous operational costs and a heavy reliance on a stable electrical grid or fossil fuels, renders them entirely inaccessible to rural farmers and environmentally detrimental. Passive solar dryers also exhibit fundamental flaws; without the integration of a thermal mass storage unit, they instantly lose their drying capability post-sunset or during overcast periods. Furthermore, their reliance on passive ventilation frequently provides insufficient airflow, causing evaporated moisture to condense on the interior glazing and drip back onto the grain, ultimately ruining the batch. Collectively, these critical limitations underscore the absolute necessity for a proposed system that intelligently integrates IOT controlled forced airflow, entirely off grid solar power, and thermal mass storage.

3.PROJECT OBJECTIVES & METHODOLOGY

3.1 OBJECTIVES

3.1.1 MAIN OBJECTIVE

The primary objective of this project is to design, develop, and evaluate a smart, 100% off grid solar powered paddy dryer. This system aims to provide a highly controlled, energy efficient, and reliable drying environment to rapidly reduce the moisture content of freshly harvested paddy, thereby preserving grain quality, increasing market value, and significantly minimizing post-harvest crop losses.

3.1.2 SPECIFIC OBJECTIVES

1.To design and optimize a high efficiency solar air heater with an enhanced internal absorber

To geometrically and thermodynamically model a solar collector featuring an optimized internal heat-exchange structure capable of maximizing the absorption of solar radiation. This collector must continuously transfer sufficient thermal energy to the forced airflow to achieve and sustain optimal drying temperatures inside the chamber, completely eliminating the need for grid-powered electrical heating elements.

2.To develop an IoT enabled smart control architecture

To geometrically and thermodynamically model a solar collector featuring an optimized internal heat-exchange structure capable of maximizing the absorption of solar radiation. This collector must continuously transfer sufficient thermal energy to the forced airflow to achieve and sustain optimal drying temperatures inside the chamber, completely eliminating the need for grid powered electrical heating elements.

3.To evaluate system performance and drying kinetics

To conduct rigorous testing protocols (including no-load system checks and load testing with wet paddy) to analyze critical performance metrics. This includes evaluating the moisture extraction rate (targeting a reduction from 25% to a safe storage level of 14%), calculating the thermal efficiency of the solar collector, and validating the system's operational reliability compared to traditional open sun drying.

3.2 METHODOLOGY

The development of the Smart Off-Grid Solar-Powered Paddy Dryer follows a systematic engineering approach divided into three critical phases: system design and simulation, component fabrication, and rigorous performance evaluation.

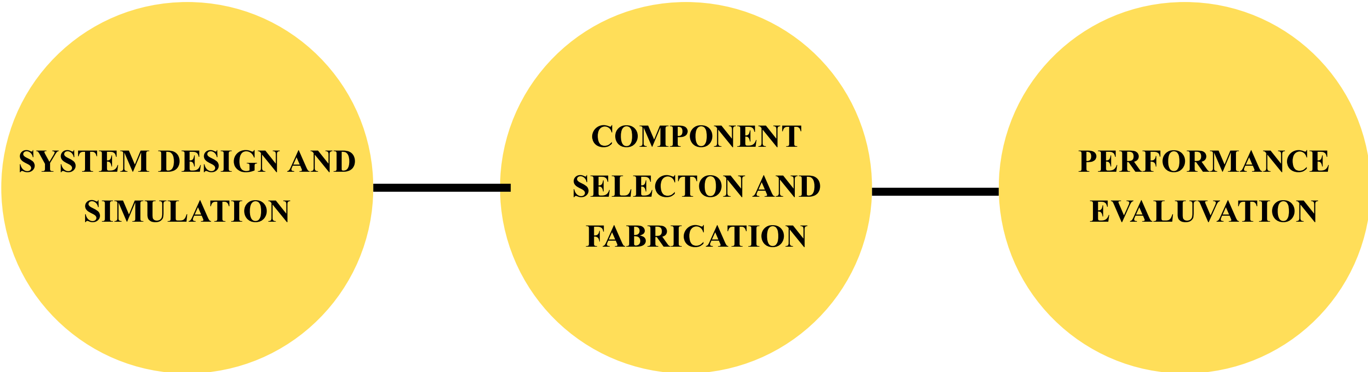

Figure 01 : Flow Diagram of Methodology

3.2.1 SYSTEM DESIGN AND SIMULATION

In the initial phase, the dryer's physical and functional architecture was established to ensure optimal thermodynamic performance.

- **CAD Modeling**

A detailed 3D model of the solar collector and drying chamber was developed using Computer Aided Design (CAD) software. This allowed for precise dimensioning of the air ducts and the drying bed to ensure structural integrity and optimal airflow distribution.

- **Thermal and Airflow Simulation**

Theoretical calculations were conducted to determine the required mass flow rate of air necessary to remove moisture effectively from a given volume of grain. The internal geometry of the solar air heater featuring an optimized absorber matrix was designed to maximize the heat trap effect. Simulation principles were applied to ensure that the forced airflow is distributed uniformly across the grain bed to prevent localized "wet spots" or overdrying.

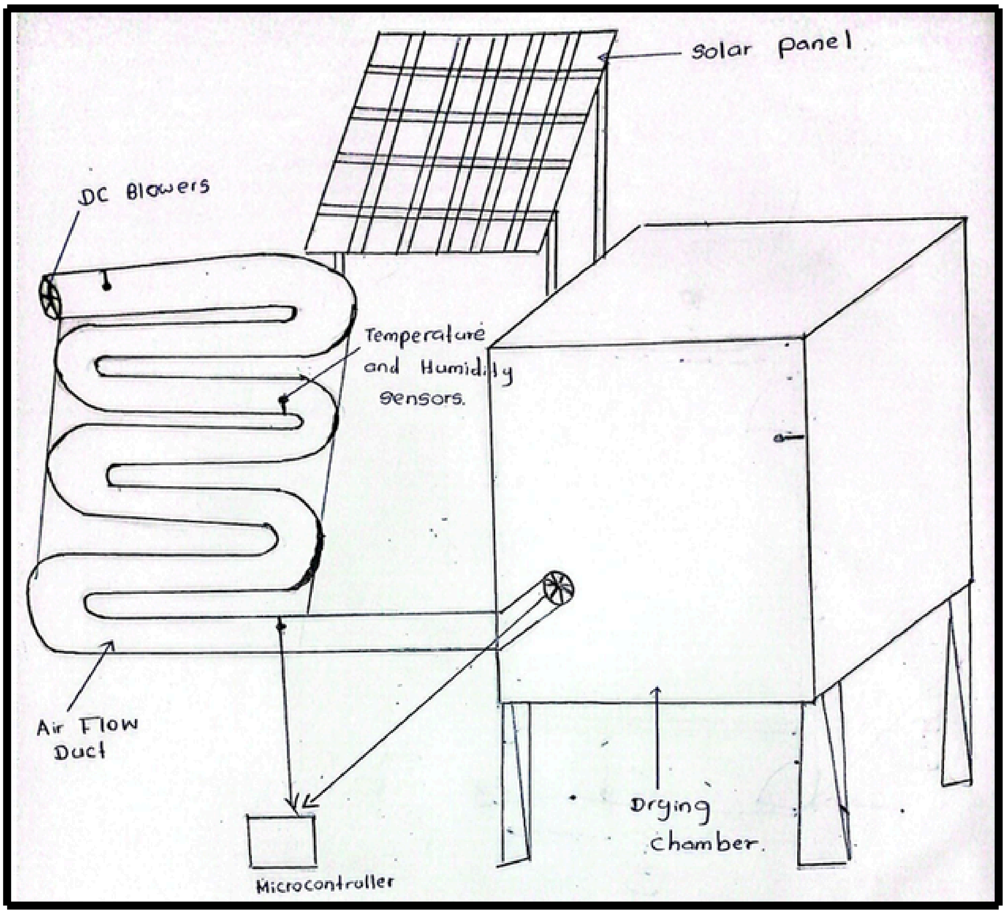

Figure 02 : Simple Sketch of System

- **Control Logic Design**

The algorithmic logic for the Microcontroller was mapped out. This includes defining the threshold values for temperature (to prevent grain cracking) and humidity (to trigger the actuated exhaust flaps) based on standard paddy drying kinetics.

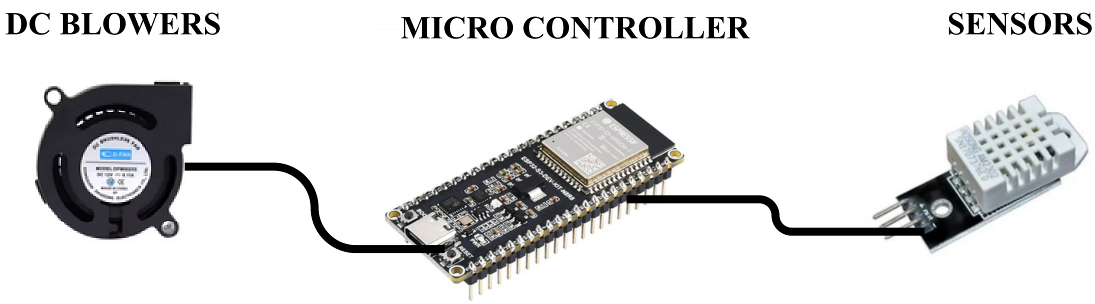

Figure 03 : Component of Control System of Project

- **Environmental Sensors**

High-precision digital temperature and humidity sensors were selected for strategic placement within the solar collector and the drying chamber. These instruments were chosen for their calibration stability and reliable performance when exposed to high-humidity agricultural environments.

- **Ventilation and Actuation Subsystems**

High-static-pressure DC blowers were chosen to overcome the physical resistance of the internal grain bed and maintain continuous forced convection. Electromechanical actuators are utilized to control the exhaust vents, allowing for precise, mechanical moisture evacuation.

- **Off-Grid Power System and Battery Pack**

The power plant consists of a solar photovoltaic panel paired with a dedicated deep cycle battery pack and an automated charge controller. Because solar irradiance fluctuates dynamically throughout the day due to changing cloud cover, a direct power connection to the solar panel would cause erratic fan speeds. The battery pack serves as a critical electrical buffer, absorbing fluctuating solar power and supplying energy uniformly to the DC blowers and control logic. This guarantees a constant, uninterrupted mass flow rate of air, which is essential for uniform drying kinetics. A step down voltage regulator is integrated into this power rail to supply a stable, low-voltage input to protect the sensitive processing circuitry.

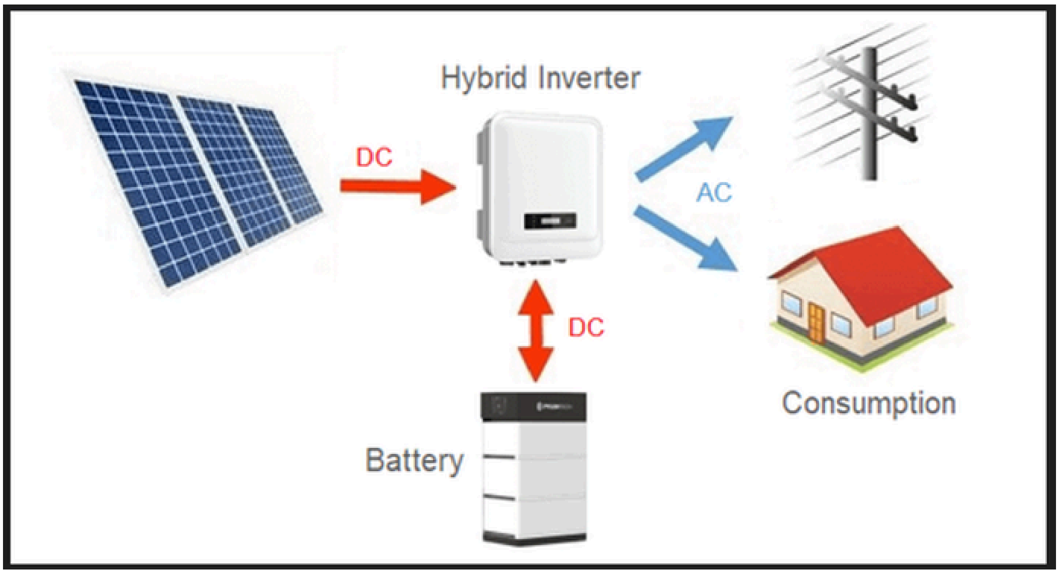

Figure 5 : Off Grid Power System and Battery Pack

3.2.2.2 FABRICATION OF THE MODEL

- **Structural Fabrication**

The structural frames of the solar air heater and the drying chamber were assembled using a rigid, heat-resistant layout. Special care was taken to ensure airtight seals along the air ducts and collector boundaries to eliminate thermal leakage. The internal absorber matrix was fixed with a calibrated air gap relative to the outer glazing to optimize the forced convective path.

- **Control and Electronic Integration**

The processing unit and its supporting power distribution networks were mounted within a centralized, weatherproof control enclosure. To isolate the delicate control circuits from the high current demanded by the ventilation system, a solid-state electronic switching interface was fabricated. This allows the controller to safely vary the fan speed using pulse width modulation without drawing high electrical currents directly through its data pins.

- **Power Subsystem Installation**

The solar photovoltaic panel was positioned at an optimal tilt angle to maximize solar energy capture. The panel was routed through the charge controller to manage the energy storage of the battery pack safely. The main system loads were drawn directly from the battery terminals, ensuring that the ventilation and voltage regulation subsystems receive uniform, ripple-free power regardless of passing clouds or transient weather variations.

- **System Assembly and Calibration**

In the final stage, the solar air heater was mechanically coupled to the drying chamber intake. The electromechanical actuators on the exhaust flaps were calibrated to ensure a reliable range of physical motion from fully closed to fully open. Finally, the control algorithm was compiled and uploaded to the embedded processor, and the wireless telemetry interface was tested to confirm real-time data transmission to the remote monitoring dashboard.

3.2.3 TESTING THE PERFORMANCE AND EVALUATION

3.2.3.1 EXPERIMENTAL SETUP AND CALIBRATION

Before data collection, all instrumentation was calibrated to ensure data integrity. High-precision digital sensors were positioned at three critical locations: the ambient air intake, the solar air heater outlet, and the center of the drying chamber. The embedded microcontroller's telemetry interface was verified to ensure that real-time data was accurately logged. The battery pack was fully charged via the solar photovoltaic panel to establish a baseline for uniform energy distribution during the testing cycle.

3.2.3.2 NO-LOAD PERFORMANCE TESTING

The primary objective of the no-load test was to determine the thermal capacity of the solar air heater and the responsiveness of the automated control logic without the influence of grain moisture.

- **Thermal Gain Analysis**

The system was operated under varying solar irradiance levels. The temperature rise ,the difference between ambient air and the air entering the chamber was measured. Due to the uniform energy supply from the battery pack, the DC blowers maintained a constant mass flow rate, allowing for stable thermal gain even during transient cloud cover.

- **Control Logic Validation**

The sensors were monitored to confirm that the electromechanical actuators triggered the exhaust vents at the 70% relative humidity threshold and that the processor successfully modulated the ventilation speed to maintain the internal temperature within the optimal 45C to 55C window.

3.2.3.3 LOAD TESTING AND DRYING KINETICS

- **Procedure**

A batch of paddy with an initial moisture content (MC) of approximately 22–25% (wet basis) was loaded into the drying chamber. The system was operated continuously during peak solar hours.

- **Data Logging**

Moisture reduction was monitored by periodic sampling and weighing of the grain. The embedded system logged internal humidity and temperature every 30 minutes, transmitting this data to the remote dashboard.

- **Efficiency of Uniform Airflow**

The stability of the battery-powered ventilation system ensured that the "drying front" moved uniformly through the grain bed, preventing the formation of wet pockets or localized over-drying that often occurs with fluctuating power sources.

3.2.3.4 LOAD TESTING AND DRYING KINETICS

- **Drying Rate**

Calculated as the total mass of moisture removed divided by the total drying time .

- **Collector Thermal Efficiency**

Evaluated by comparing the heat energy absorbed by the airflow to the total solar radiation incident on the collector glazing.

- **Specific Energy Consumption**

An analysis of the battery pack's performance to ensure that the energy utilized by the fans and control logic remained within the sustainable limits of the solar-recharging cycle.

- **Grain Quality Assessment**

Post-drying analysis focused on the "Head Rice Yield" (HRY), ensuring that the automated temperature regulation prevented thermal stress and minimized grain cracking compared to traditional open-sun drying methods.

3.3 SOFTWARES

Table 01 : Software Requirements and Acquisition Plan for Smart Solar Powered Paddy Dryer

Software	Purpose	Plan to Acquire
SolidWorks	3D modeling and design of housing and frame	Already available
ANSYS	simulation for airflow and heat exchanger	Use ANSYS Student version
Proteus	Circuit simulation	Free version
Arduino IDE	Programming the microcontroller	Free download from official site
Microsoft Excel	Data analysis and documentation	Already available
MS Word	Report writing and presentation	Already available
MATLAB	Additional data analysis, modeling, or control simulation	Student license

4.PROJECT PLANNING AND TIMELINE

4.1 MILESTONES

- **Research and Conceptualization**

In this phase, we studied existing paddy drying methods, identified their limitations, and analyzed the need for a solar-powered smart dryer. Literature was reviewed, problems were identified, and a suitable concept was developed to achieve efficient, low-cost, and uniform drying.

- **Detailed Design and Analysis by Calculation**

in this phase, the system requirements were finalized. The solar collector and drying chamber were designed in detail. Necessary calculations such as thermal analysis, airflow analysis, and structural calculations were performed. Components were selected based on the design.

- **Simulation and Verification**

In this phase, 3D models were created using SolidWorks and thermal as well as airflow simulations were carried out. The design was verified for performance. A model was developed and tested under no-load and load conditions. Sensors were calibrated and system functionality was verified.

- **Final Evaluation and Documentation**

In the final phase, the system performance was evaluated. Moisture reduction, drying efficiency, airflow, and temperature results were analyzed and compared with traditional drying methods. Finally, the report was prepared and the documentation completed.

4.2 TIME FRAME

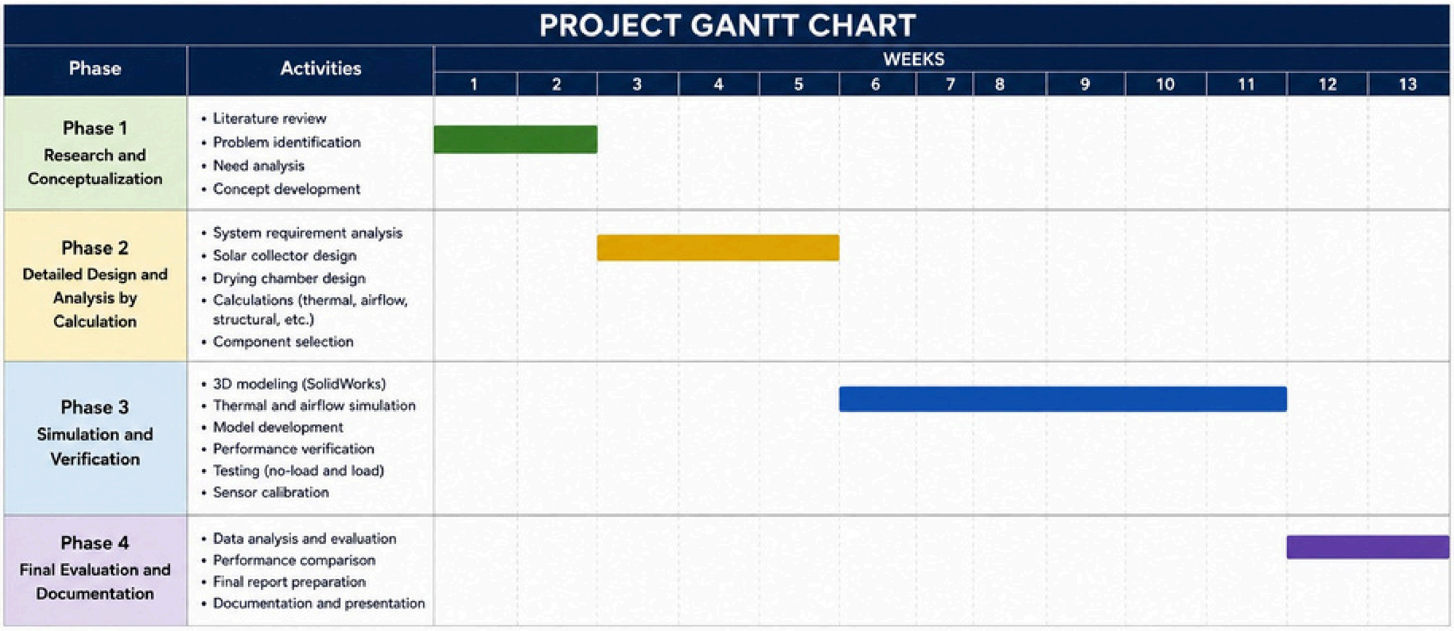

Figure 6 : Project Gantt Chart

5. IMPACT ANALYSIS

- ### Economic Impact

From an economic perspective, the system drastically reduces operational costs by substituting traditional, expensive fossil fuels or high-tariff grid electricity with free, renewable solar energy. By ensuring uniform and controlled drying, the smart dryer minimizes post-harvest losses caused by spoilage, over-drying, or unpredictable weather. This directly translates to higher-quality grain yields, allowing farmers to command premium market prices, recover their initial capital investment quickly, and secure more stable profit margins.

- ### Social Impact

Socially, this technology elevates the quality of life for farming communities by modernizing labor-intensive agricultural workflows. Automating the monitoring and drying processes reduces the physical burden and time commitment typically required by open-sun drying methods, which demand constant manual turning and guarding against sudden rainfall. Furthermore, by improving food security through reduced crop wastage and ensuring safer, mold-free grain production, the project supports healthier dietary outcomes and strengthens the resilience of rural agricultural economies.

- **Environmental Impact**

Environmentally, the smart paddy dryer offers a clean, zero-emission alternative to conventional mechanical dryers that rely on diesel or biomass burning. By utilizing solar thermal energy and optimizing airflow through smart automation, the system substantially lowers the carbon footprint of the post-harvest supply chain. Additionally, transitioning away from traditional open-yard sun drying prevents localized dust contamination and reduces the reliance on burning agricultural waste for heat, thereby contributing to cleaner air and a healthier local ecosystem.

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